1.9.3 Expanded form of numbers using exponents

In the lower classes, we have learnt how to write a whole number in the expanded form.
For example, $5832 = 5\times1000 + 8\times100 + 3\times10 + 2\times1$

$= 5\times10^3 + 8\times10^2 + 3\times10^1 + 2$ (when we use exponential notation).

What shall we do if we get decimal places? Powers of 10 with negative exponents come to our rescue!

Thus, $58.32 = 50 + 8 + \frac{3}{10} + \frac{2}{100}$

$= 5\times10 + 8\times1 + 3\times\frac{1}{10} + 2\times\frac{1}{100}$

$= 5\times10^1 + 8\times10^0 + 3\times10^{-1} + 2\times10^{-2}$



Try these

Expand the following numbers using exponents:

1. 8120
2. 20305
3. 3652.01
4. 9426.521

1.9.4 Laws of Exponents

Laws of exponents arise out of certain basic ideas. A **positive exponent** of a number indicate how many times we use that number in a multiplication whereas a **negative exponent** suggests us how many times we use that number in a division, since the opposite of multiplying is dividing.

- **Product law**

According to this law, when multiplying two powers that have the same base, we can add the exponents. That is,

$$a^m \times a^n = a^{m+n}$$

where a ($a\neq0$), m, n are integers. Note that the base should be the same in both the quantities.

Examples:

a)	$2^3\times2^2 = 2^5$ by law(meaning $8\times4 = 32$ and note that it is not $2^{3\times2}$)
b)	$(-2)^{-4}\times(-2)^{-3} = (-2)^{(-4)+(-3)}$ by law and so $= (-2)^{-7}$ (or) $(-2)^{-4}\times(-2)^{-3} = \frac{1}{(-2)^4}\times\frac{1}{(-2)^3} = \frac{1}{(-2)^4\times(-2)^3} = \frac{1}{(-2)^{4+3}} = \frac{1}{(-2)^7} = (-2)^{-7}$
c)	$(-5)^3\times(-5)^{-3} = (-5)^{3-3}$ by law and so $= (-5)^0 = 1$

- **Quotient law**

According to this law, when dividing two powers that have the same base we can subtract the exponents. That is,

$$\frac{a^m}{a^n} = a^{m-n}$$

where a ($a \neq 0$), m, n are integers. Note that the base should be the same in both the quantities. How does it work? Study the following examples.

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